A062 - SST Microgravity Source Falsifier

v0.4.0: Finite Vortex-Tube Energy, Organized Swirl Transport and Gravitational Closure

SST-Workbench – source-pinned, CPU-reproducible

13 October 2026

1 Identification and scope

The present instance is an extension of, not a replacement for, A062-v0.1.0 to v0.3.0. Competing hypotheses are isolated: a finite-width Gaussian-vorticity tube with finite kinetic energy; the localized stress double-divergence gravity candidate in SST-73; and a separate postulated bulk gravity/clock mode in SST Canon v0.8.35. A positive energy is not a derivation of a gravitational force.

Automatic run provenance

Field	Recorded value
Run ID	20261013T155937677807Z-6267ca54
Run phase	BLIND
Protocol bundle SHA-256	df9816159083c762878493d940fd78c6fcb8893d8889296001b5cfbaf123dfbb
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	596b8804619e3be5a715ffc35b4df79f80306d512bc97e4b1b8d27a3c3b1efb3

Gate ledger

Gate	Status	Question
G0	PASS	Did framework v1.0.8 freeze and validate protocol, provenance and local integrity?
G1	PASS	Are E011 Gilbert and KnotPlot independent original geometry SHA256 verified?
G2	PASS	Is Gaussian finite-width vortex tube energy convergent, positive and vorticity solenoidal?
G3	PASS	Does integrated kinetic energy yield a finite nonzero energy-equivalent mass candidate wit
G4	FAIL	Does SST-73 organized stress double-divergence produce a free-space monopole for localiz
G5	PASS	Does independent C++17 FP64 reproduce normalized finite Gaussian vortex-tube energy
G6	NOT_APPLICABLE	Is time-dependent unforced Euler/Navier-Stokes finite-core stability and stress balance qu
G7	PASS	Do upstream-independent E011 geometries reproduce normalized tube energy within pred
G8	UNRESOLVED	Are physical clock mediator, universal mass coupling and dimensional Newton-G intrinsic
G9	DEFERRED	Was reveal separately authorized and executed?

Backend manifest


```
{
  "cpp_fp64": {
    "comparisons": [
      {
        "cpp_I": 1.6240955321895743,
        "provider": "gilbert",
        "python_I": 1.6240955321895743,
        "relative_error": 0.0
      },
      {
```

```
        "cpp_I": 1.8470697062769819,
        "provider": "knotplot",
        "python_I": 1.8470697062769816,
        "relative_error": 1.202145236698144e-16
      }
    ],
    "compiler": "C:\\workspace\\Strawberry\\c\\bin\\g++.EXE",
    "max_relative_error": 1.202145236698144e-16,
    "status": "PASS"
  },
  "limitations": [
    "E011 STATIC_READY does not imply DYNAMICS_READY",
    "No Navier-Stokes time integration",
    "No gravitation closure"
  ],
  "python_reference": {
    "actual_backend": "python-numpy-fp64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
  "schema": "SST-BACKEND-MANIFEST-2",
  "scientific_boundary": "Gauss vortex-tube energy is genuinely nonzero; identification with universal
  "sycl_gpu": {
    "authority": "SCREENING_ONLY",
    "requested": false
  }
}
```


Source manifest


```
{
  "manifest_sha256": "1689c7aeae60c37713a8e2554d75b2aa916e454ffe21f399bb21306e3f02b726",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "E011",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "gilbert",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A062_SST_Microgravity
      "provenance_family": "gilbert_ideal_A004",
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "discovery",
      "sha256": "274a4ce111607a6295e280f3f8369c665e7de82ac33afcd4cb227eb61568d530",
      "source_id": "S_GILBERT",
      "version": "0.3.0"
    },
    {
      "catalog_id": "E011",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "knotplot",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A062_SST_Microgravity
      "provenance_family": "knotplot_relaxed_A001",
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
```

```
      "role": "confirmation",
      "sha256": "8fea8a15ff3047dbbfb4aba05090b9a6cadc10eeedf97c22350925c764bc540e",
      "source_id": "S_KNOTPLOT",
      "version": "0.3.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


2 Source provenance and prerequisites

Both E011/SKLSA v0.3.0 STATIC_READY trefoil carriers are separately authenticated by raw SHA-256: Gilbert ideal Fourier record 3:1:1 and KnotPlot VECT coarse relaxed knot. `STATIC_READY` does not mean `DYNAMICS_READY`. Opposite signed writhe indications are not erased. See `inputs/SST_KNOWLEDGE_CENSUS.json` for paper-scoped literature provenance; unread articles are not silently used as evidence.

3 Resolved smooth vorticity-tube candidate

For each closed smooth or piecewise-smooth curve γ and Gaussian width $a > 0$, define

$$\boldsymbol{\omega}_a(\mathbf{x}) = \Gamma \oint_\gamma d\mathbf{X}\, \frac{\exp[-|\mathbf{x}-\mathbf{X}|^2/(2a^2)]}{(2\pi a^2)^{3/2}}. \tag{1}$$

This is a three-dimensional smooth, finite-width vorticity distribution; its divergence vanishes exactly for a closed curve because the derivative along the curve is a total derivative. It need not be a stationary Euler or Navier–Stokes solution.

The compatible free-space kinetic energy for constant ambient density is

$$E_a = \frac{\rho_f \Gamma^2}{8\pi} \oint_\gamma \oint_\gamma \frac{\operatorname{erf}(|\mathbf{X}-\mathbf{Y}|/(2a))}{|\mathbf{X}-\mathbf{Y}|}\, d\mathbf{X}\cdot d\mathbf{Y}. \tag{2}$$

The finite diagonal limit is $1/(a\sqrt{\pi})$. This kernel avoids arbitrary grid-cell cutoff and supports independent C++/Python quadrature. Only ambient density ρ_f enters E_a. Substituting the much larger ρ_{core} into the whole bath is *not a valid physical inference.*

For a diagnostic physical length, the code reports $L_{\text{phys}} = r_c/a_{\text{normalized}}$; this *assumes* $a_{\text{core}} = r_c$, whereas SST Canon v0.8.35 distinguishes the two. SI energy and mass under that scale hypothesis must not be reported as a derived electron mass.

4 Three candidate gravitational source routes

The SST-73 organized transport tensor is $\Sigma_{ij} = \rho_f u_i u_j$. Its preferred source $S_C = \lambda \partial_i \partial_j \Sigma_{ij}$ obeys

$$\int_{\mathbb{R}^3} S_C\, d^3x = \lambda \lim_{R\to\infty} \oint_{S_R} n_i\, \partial_j \Sigma_{ij}\, dA = 0 \tag{3}$$

under the stated free-space decay assumptions. Changing from pressure to a stress double divergence does not change this obstruction for a localized constant-density Euler velocity field. Non-decaying envelopes, boundary sources and nonlocal constitutive responses are not excluded.

The candidate energy source is instead

$$\rho_E(\mathbf{x}) = \tfrac{1}{2}\rho_f |\mathbf{u}(\mathbf{x})|^2, \qquad Q_E = \frac{1}{c^2}\int \rho_E\, d^3x > 0. \tag{4}$$

This follows as *energy-equivalent mass* from a nontrivial tube, not as a theorem about active gravity. A separate *postulate* $\nabla^2\Phi = 4\pi K \rho_E/c^2$ would generate an inverse-square far field only because that mediator and charge coupling were introduced explicitly. Deriving K, universal attraction and the same coupling to a test knot remain open.

5 Viscosity, Euler and Navier–Stokes

For a smooth unforced incompressible Navier–Stokes flow with sufficient decay,

$$\frac{dE}{dt} = -\rho_f \nu \int |\boldsymbol{\omega}|^2 \, d^3x. \tag{5}$$

The code computes the integral analytically from a Gaussian double-line convolution, returning the coefficient of an *optional* viscosity. This is a screen, not numerical evolution, a dynamical stability claim, or a use of an alleged solved Millennium Problem. Euler is $\nu = 0$ in this specific dissipation identity, without guaranteeing stationarity.

6 Predeclared numerical gates

The maximum last-resolution change for the energy integral must be below 0.007; the scaled quadrature divergence below 10^{-5}; normalized provider integral disagreement below 0.20; and independent native FP64 relative error below 2×10^{-10}. G4 may FAIL while later gates run. G6 and G8 remain UNRESOLVED until true dynamic and scalar-gravity qualifications exist. Conditional comparisons do not retroactively redefine the frozen criteria.

7 Scientific interpretation

A nonzero kinetic energy is a source of nonzero energy-equivalent mass. For isolated locally smooth stress of the SST-73 double-divergence form, the integrated gravity monopole remains zero. A universal gravitational force has therefore neither been derived nor ruled out in every SST variant; the tested local stress-source hypothesis fails, while the energy-source hypothesis is mathematically admissible but constitutively and dynamically underdetermined.

8 Blindness and reproducibility

Framework v1.0.8 `LOCAL_VALIDATED_CANDIDATE` is bundled unmodified. The instance FREEZE launcher creates 32-byte private nonce/key files only when absent and before any commitments/freeze, never rotates existing files; private secrets are excluded from source ZIP. Unique run IDs prevent rerun overwrite. Public literature and benchmarks do not constitute held-out experimental validation.

References

[1] SST-18, *Unifying Electromagnetism, Gravity and Quantum Geometry via Incompressible Hydrodynamics.*

[2] SST-34, *SST-Hydrogen: Short-Range Circulation Coupling (Deprecated as Gravity).*

[3] SST-59, *Mass Hydrodynamic Topology.*

[4] SST-61, *Helicity-Constrained Stability Beyond Energy Minimization.*

[5] SST-73, *A Poisson-Type Gravity Program from Organized Swirl Transport*, esp. Secs. 9, 13, 16, Appendix B.

[6] SST Canon v0.8.35, esp. Secs. 11.11, 11.16–11.17; explicit calibrated-G guard.

[7] A062-v0.1.0 to v0.3.0 scientific gate reports, archived independently.